Exam 1 Questions (2.A – 5.B)

CHEM 1211K | Fall 2025

QUESTION 1 (5 points)

1. Thorium-232 (^{232}Th) is a radioactive isotope of thorium. When it decays, it emits a helium nucleus (^{4}He). What is the name of the element left behind after this decay? **Note:** the total numbers of protons and neutrons are conserved in this process.

   $^{232}Th \rightarrow {}^{4}He + ?$

   A. ◯ **Radium**
   B. ◯ Uranium
   C. ◯ Thorium
   D. ◯ Protactinium
   E. ◯ Actinium

2. Radium-224 (^{224}Ra) is a radioactive isotope of radium. When it decays, it emits a helium nucleus (^{4}He). What is the name of the element left behind after this decay? **Note:** the total numbers of protons and neutrons are conserved in this process.

   $^{224}Ra \rightarrow {}^{4}He + ?$

   A. ◯ **Radon**
   B. ◯ Radium
   C. ◯ Francium
   D. ◯ Thorium
   E. ◯ Actinium

3. Polonium-212 (^{212}Po) is a radioactive isotope of polonium. When it decays, it emits a helium nucleus (^{4}He). What is the name of the element left behind after this decay? **Note:** the total numbers of protons and neutrons are conserved in this process.

   $^{212}Po \rightarrow {}^{4}He + ?$

   A. ◯ **Lead**
   B. ◯ Radon
   C. ◯ Astatine
   D. ◯ Mercury
   E. ◯ Radium

QUESTION 2 (5 points)

4. What is the chemical formula for the ionic compound that contains the cobalt(II) cation and nitrite anion?

   A. ◯ **$Co(NO_2)_2$**
   B. ◯ $Co(NO_2)_3$
   C. ◯ Co_2NO_2
   D. ◯ $Co(NO_3)_2$
   E. ◯ $CoNO_2$

5. What is the chemical formula for the ionic compound that contains the ammonium cation and sulfate anion?

   A. ◯ **$(NH_4)_2SO_4$**
   B. ◯ NH_4SO_4
   C. ◯ $NH_4(SO_4)_2$
   D. ◯ $(NH_4)_2SO_3$
   E. ◯ $NH_4(SO_3)_2$

6. What is the chemical formula for the ionic compound that contains the lithium cation and phosphate anion?

   A. ◯ **Li_3PO_4**
   B. ◯ $Li(PO_4)_3$
   C. ◯ $LiPO_3$
   D. ◯ Li_3PO_3
   E. ◯ $LiPO_4$

QUESTION 3 (5 points)

7. An unknown element has two isotopes with masses 36.97 amu (isotope **A**) and 38.95 amu (isotope **B**). If a sample of this element contains 68.2% **A** and 31.8% **B**, what is the average atomic mass of the element? Enter your response in atomic mass units (amu) to the nearest 0.1 amu.

   37.6 amu

8. An unknown element has two isotopes with masses 36.97 amu (isotope **A**) and 38.95 amu (isotope **B**). If a sample of this element contains 25.6% **A** and 74.4% **B**, what is the average atomic mass of the element? Enter your response in atomic mass units (amu) to the nearest 0.1 amu.

   38.4 amu

9. An unknown element has two isotopes with masses 36.97 amu (isotope **A**) and 38.95 amu (isotope **B**). If a sample of this element contains 52.1% **A** and 47.9 % **B**, what is the average atomic mass of the element? Enter your response in atomic mass units (amu) to the nearest 0.1 amu.

   37.9 amu

QUESTION 4 (6 points)

10. Match each count of protons and electrons with the corresponding atom or ion.

30 protons and 28 electrons	$\mathbf{Zn^{2+}}$
16 protons and 18 electrons	$\mathbf{S^{2-}}$
12 protons and 12 electrons	**Mg**

11. Match each count of protons and electrons with the corresponding atom or ion.

25 protons and 22 electrons	$\mathbf{Mn^{3+}}$
9 protons and 10 electrons	$\mathbf{F^{-}}$
32 protons and 32 electrons	**Ge**

12. Match each count of protons and electrons with the corresponding atom or ion.

29 protons and 27 electrons	$\mathbf{Cu^{2+}}$
53 protons and 54 electrons	$\mathbf{I^{-}}$
19 protons and 19 electrons	**K**

QUESTION 5 (4 points)

13. What is the formula mass of acetohydroxamic acid, which has a molecular formula of $C_2H_5NO_2$?

   A. ◯ **75.1 amu**
   B. ◯ 43.0 amu
   C. ◯ 59.1 amu
   D. ◯ 71.0 amu
   E. ◯ 63.1 amu

14. What is the formula mass of salicylhydroxamic acid, which has a molecular formula of $C_7H_7NO_3$?

   A. ◯ **153.1 amu**
   B. ◯ 75.1 amu
   C. ◯ 121.1 amu
   D. ◯ 147.1 amu
   E. ◯ 43.0 amu

15. What is the formula mass of glucosamine, which has a molecular formula of $C_6H_{13}NO_5$?

   A. ◯ **179.2 amu**
   B. ◯ 43.0 amu
   C. ◯ 115.2 amu
   D. ◯ 165.2 amu
   E. ◯ 119.1 amu

QUESTION 6 (5 points)

16. The compound H_2CrO_4 has the name chromic acid. What is the name of the CrO_3^{2-} anion?

   A. ◯ **Chromite**
   B. ◯ Chromate
   C. ◯ Perchromate
   D. ◯ Hypochromite
   E. ◯ Chromium trioxide

17. The compound H_2SeO_4 has the name selenic acid. What is the name of the SeO_3^{2-} anion?

   A. ◯ **Selenite**
   B. ◯ Selenate
   C. ◯ Perselenate
   D. ◯ Hyposelenite
   E. ◯ Selenium trioxide

18. The compound H_2CrO_4 has the name chromic acid. What is the name of the CrO_4^{2-} anion?

   A. ◯ **Chromate**
   B. ◯ Hypochromite
   C. ◯ Chromite
   D. ◯ Perchromate
   E. ◯ Chromium trioxide

QUESTION 7 (5 points)

19. The molar mass of sodium dichromate ($Na_2Cr_2O_7$) is 262.0 g·mol^{-1}. What mass of chromium is present in a sample of this compound weighing 120.0 g? Enter your response in grams (g) to the nearest 0.1 g.

   47.6 g

20. The molar mass of sodium chromate (Na_2CrO_4) is 162.0 g·mol^{-1}. What mass of chromium is present in a sample of this compound weighing 125.0 g? Enter your response in grams (g) to the nearest 0.1 g.

   40.1 g

QUESTION 8 (5 points)

21. P_4, known as "white phosphorus," is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

    [S1] The bonds between phosphorus atoms are covalent. **True**
    [S2] The average mass of a molecule of P_4 is 30.97 amu. **False**
    [S3] The empirical formula of white phosphorus is P. **True**
    [S4] There are four times as many phosphorus atoms as P_4 molecules in any sample of white phosphorus. **True**
    [S5] P_4 is best described as a compound because it contains molecules rather than atoms. **False**

22. P_4, known as "white phosphorus," is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

    [S1] The bonds between phosphorus atoms are ionic. **False**
    [S2] The average mass of a molecule of P_4 is 123.9 amu. **True**
    [S3] The empirical formula of white phosphorus is P. **True**
    [S4] There are four times as many P_4 molecules as phosphorus atoms in any sample of white phosphorus. **False**
    [S5] P_4 is best described as a compound because it contains molecules rather than atoms. **False**

23. P_4, known as "white phosphorus," is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

    [S1] The bonds between phosphorus atoms are covalent. **True**
    [S2] The average mass of a molecule of P_4 is 123.9 amu. **True**
    [S3] The empirical formula of white phosphorus is P_4. **False**
    [S4] There are four times as many phosphorus atoms as P_4 molecules in any sample of white phosphorus. **True**
    [S5] P_4 is best described as an element because it contains only one type of atom. **True**

QUESTION 9 (5 points)

24. In an experiment involving the combination of magnesium hydroxide and phosphoric acid, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?

3 $Mg(OH)_2$ + **2** H_3PO_4 → **6** H_2O + ?

A. ◯ **$Mg_3(PO_4)_2$**
B. ◯ $Mg_2(PO_4)_3$
C. ◯ $MgPO_4$
D. ◯ Mg_3PO_4
E. ◯ $Mg(PO_4)_2$

25. In an experiment involving the combination of aluminum and sulfur, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?

16 Al + **3** S_8 → **8** ?

A. ◯ **Al_2S_3**
B. ◯ Al_3S_2
C. ◯ AlS
D. ◯ Al_3S
E. ◯ AlS_2

26. In an experiment involving the combination of calcium chloride and sodium phosphate, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?

3 $CaCl_2$ + **2** Na_3PO_4 → **6** NaCl + **1** ?

A. ◯ **$Ca_3(PO_4)_2$**
B. ◯ $Ca_2(PO_4)_3$
C. ◯ $CaPO_4$
D. ◯ Ca_3PO_4
E. ◯ $Ca(PO_4)_2$

QUESTION 10 (5 points)

27. A 100.0-gram sample of an unknown compound contains 52.2 g carbon, 4.4 g hydrogen, 20.3 g nitrogen, and 23.1 g oxygen. What is the empirical formula of this compound?

A. ◯ **C_3H_3NO**
B. ◯ $C_4H_6N_2O$
C. ◯ C_4H_4NO
D. ◯ C_3H_7NO
E. ◯ $C_4H_{11}NO_2$

28. A 100.0-gram sample of an unknown compound contains 70.6 g carbon, 4.2 g hydrogen, 11.8 g nitrogen, and 13.4 g oxygen. What is the empirical formula of this compound?

A. ◯ **C_7H_5NO**
B. ◯ C_3H_3NO
C. ◯ C_6H_7NO
D. ◯ C_7H_9NO
E. ◯ $C_7H_{11}NO_2$

QUESTION 11 (6 points)

29. Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.

[R1] $3\ Li(s) + AlCl_3(s) \rightarrow 3\ LiCl(s) + Al(s)$ **single replacement**
[R2] $C_2H_6(g) + 3.5\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(g)$ **combustion**
[R3] $4\ Na(s) + O_2(g) \rightarrow 2\ Na_2O(s)$ **synthesis**

30. Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.

[R1] $H_3PO_4(aq) + 3\ KOH(aq) \rightarrow K_3PO_4(aq) + 3\ H_2O(l)$ **double replacement**
[R2] $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ **decomposition**
[R3] $2\ N_2(g) + 5\ O_2(g) \rightarrow 2\ N_2O_5(g)$ **synthesis**

31. Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.

[R1] $3\ H_2SO_4(aq) + 2\ Al(OH)_3(s) \rightarrow Al_2(SO_4)_3(aq) + 6\ H_2O(l)$ **double replacement**
[R2] $Zn(s) + 2\ AgNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + 2\ Ag(s)$ **single replacement**
[R3] $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ **synthesis**

QUESTION 12 (4 points)

32. What aqueous species are produced, and in what amounts, when 1 mole of $NaNO_3$ is dissolved in water?

A. ◯ **1 mol Na^+(aq) and 1 mol NO_3^-(aq)**
B. ◯ 1 mol Na^+, 1 mol N^{5+}(aq), and 3 mol O^{2-}(aq)
C. ◯ 1 mol Na^+(aq), 0.5 mol N_2(aq), and 1.5 mol O_2(aq)
D. ◯ 1 mol Na(aq) and 1 mol NO_3(aq)
E. ◯ 1 mol $NaNO_3$(aq); this compound does not dissociate in solution.

33. What aqueous species are produced, and in what amounts, when 1 mole of $NaBrO_3$ is dissolved in water?

A. ◯ **1 mol Na^+(aq) and 1 mol BrO_3^-(aq)**
B. ◯ 1 mol Na^+(aq), 1 mol Br^{5+}(aq), and 3 mol O^{2-}(aq)
C. ◯ 1 mol Na^+(aq), 1 mol Br^-(aq), and 3 mol O(aq)
D. ◯ 1 mol Na^+(aq), 1 mol Br^-(aq), and 1.5 mol O_2(aq)
E. ◯ 1 mol $NaBrO_3$(aq); this compound does not dissociate in solution.

34. What aqueous species are produced, and in what amounts, when 1 mole of $NaClO_4$ is dissolved in water?

A. ◯ **1 mol Na^+(aq) and 1 mol ClO_4^-(aq)**
B. ◯ 1 mol Na(aq) and 1 mol ClO_4(aq)
C. ◯ 1 mol Na^+(aq), 1 mol Cl^{7+}(aq), and 4 mol O^{2-}(aq)
D. ◯ 1 mol Na^+(aq) and 1 mol ClO_4(aq)
E. ◯ 1 mol $NaClO_4$(aq); this compound does not dissociate in solution.

QUESTION 13 (5 points)

35. Which of the following combinations of ionic solutions would *NOT* result in the formation of a precipitate?

A. ◯ **$Na_2SO_4(aq) + KNO_3(aq)$**
B. ◯ $LiCl(aq) + AgNO_3(aq)$
C. ◯ $CoCl_2(aq) + NaOH(aq)$
D. ◯ $Na_3PO_4(aq) + CaBr_2(aq)$
E. ◯ $Fe(NO_3)_3(aq) + K_2CO_3(aq)$

36. Which of the following combinations of ionic solutions would *NOT* result in the formation of a precipitate?

A. ◯ **$NaBr(aq) + K_2CO_3(aq)$**
B. ◯ $CoCl_2(aq) + NaOH(aq)$
C. ◯ $Na_3PO_4(aq) + CaBr_2(aq)$
D. ◯ $Pb(NO_3)_2(aq) + Na_2CrO_4(aq)$
E. ◯ $BaCl_2(aq) + Na_2SO_4(aq)$

37. Which of the following combinations of ionic solutions would *NOT* result in the formation of a precipitate?

A. ◯ **$KNO_3(aq) + Li_2SO_4(aq)$**
B. ◯ $NaI(aq) + Pb(NO_3)_2(aq)$
C. ◯ $Na_2S(aq) + CdCl_2(aq)$
D. ◯ $CoCl_2(aq) + NaOH(aq)$
E. ◯ $BaCl_2(aq) + Na_2SO_4(aq)$

QUESTION 14 (5 points)

38. What mass of carbon dioxide (CO_2, 44.0 g/mol) is produced when 10.5 kg of propane (C_3H_8, 44.1 g/mol) is fully combusted according to the balanced reaction below? Assume that the reaction goes to completion.

    $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(g)$

    A. ◯ **31.4 kg**
    B. ◯ 33.8 kg
    C. ◯ 11.3 kg
    D. ◯ 462 kg
    E. ◯ 10.5 kg

39. What mass of water (H_2O, 18.0 g/mol) is produced when 10.5 kg of propane (C_3H_8, 44.1 g/mol) is fully combusted according to the balanced reaction below? Assume that the reaction goes to completion.

    $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(g)$

    A. ◯ **17.1 kg**
    B. ◯ 4.61 kg
    C. ◯ 189 kg
    D. ◯ 10.5 kg
    E. ◯ 1.02 kg

QUESTION 15 (5 points)

40. Classify each aqueous solution as a non-electrolyte, weak electrolyte, or strong electrolyte.

[S1] 0.10 M $HF(aq)$	**weak electrolyte**
[S2] 0.10 M $C_{12}H_{22}O_{11}(aq)$ (aqueous sucrose)	**non-electrolyte**
[S3] 0.10 M $LiNO_3(aq)$	**strong electrolyte**

41. Classify each aqueous solution as a non-electrolyte, weak electrolyte, or strong electrolyte.

[S1] 0.10 M $CH_3OH(aq)$ (aqueous ethanol)	**non-electrolyte**
[S2] 0.10 M $KBr(aq)$	**strong electrolyte**

[S3] 0.10 M $HNO_3(aq)$ **strong electrolyte**

42. Classify each aqueous solution as a non-electrolyte, weak electrolyte, or strong electrolyte.

[S1] 0.10 M $HCN(aq)$ **weak electrolyte**
[S2] 0.10 M $C_{12}H_{22}O_{11}(aq)$ (aqueous sucrose) **non-electrolyte**
[S3] 0.10 M $HBr(aq)$ **strong electrolyte**

QUESTION 16 (5 points)

43. The compound P_4S_{10} reacts vigorously with water to form phosphoric acid and hydrogen sulfide according to the balanced reaction below.

$P_4S_{10} + 16\ H_2O \rightarrow 4\ H_3PO_4 + 10\ H_2S$

How many moles of water are needed to synthesize 33.1 moles of H_2S? Assume that an excess of P_4S_{10} is available.

A. ◯ **53.0 mol**
B. ◯ 20.7 mol
C. ◯ 33.1 mol
D. ◯ 0.0483 mol
E. ◯ 16.0 mol

44. The compound P_4S_{10} reacts vigorously with water to form phosphoric acid and hydrogen sulfide according to the balanced reaction below.

$P_4S_{10} + 16\ H_2O \rightarrow 4\ H_3PO_4 + 10\ H_2S$

How many moles of water are needed to synthesize 33.1 moles of H_3PO_4? Assume that an excess of P_4S_{10} is available.

A. ◯ **132 mol**
B. ◯ 8.23 mol
C. ◯ 33.1 mol
D. ◯ 16.0 mol
E. ◯ 0.121 mol

QUESTION 17 (5 points)

45. A partially characterized compound **A**Cl_2 is a gas that dissolves in water but does not dissociate into ions. Which of the following elements is most likely the identity of **A**?

A. ◯ **Oxygen (O)**
B. ◯ Magnesium (Mg)
C. ◯ Sodium (Na)
D. ◯ Barium (Ba)
E. ◯ Nickel (Ni)

46. A partially characterized compound **A**Cl_2 is a liquid that is insoluble in water. When it is dissolved in the organic solvent dichloromethane, it does not dissociate into ions. Which of the following elements is most likely the identity of **A**?

A. ◯ **Sulfur (S)**
B. ◯ Magnesium (Mg)
C. ◯ Sodium (Na)
D. ◯ Barium (Ba)
E. ◯ Nickel (Ni)

QUESTION 18 (5 points)

47. Which of the following is the correct balanced net ionic equation for the single replacement reaction of zinc metal with hydrobromic acid?

A. ◯ **$Zn(s) + 2\ H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$**
B. ◯ $Zn(s) + 2\ HBr(aq) \rightarrow ZnBr_2(aq) + H_2(g)$
C. ◯ $Zn(s) + 2\ H^+ + 2\ Br^-(aq) \rightarrow Zn^{2+}(aq) + 2\ Br^-(aq) + H_2(g)$
D. ◯ $Zn(s) + HBr(aq) \rightarrow ZnBr_2(aq) + H_2(g)$
E. ◯ $Zn(s) + H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

48. Which of the following is the correct balanced net ionic equation for the single replacement reaction of magnesium metal with hydrochloric acid?

A. ◯ **$Mg(s) + 2\ H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$**
B. ◯ $Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
C. ◯ $Mg(s) + 2\ H^+ + 2\ Cl^-(aq) \rightarrow Mg^{2+}(aq) + 2\ Cl^-(aq) + H_2(g)$
D. ◯ $Mg(s) + HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

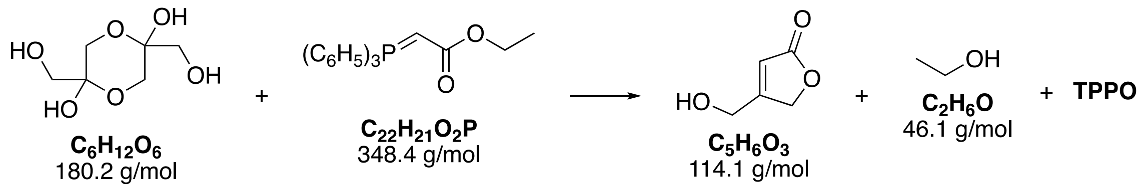
E. ◯ $Mg(s) + H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$

49. Which of the following is the correct balanced net ionic equation for the single replacement reaction of magnesium metal with nitric acid?

A. ◯ **$Mg(s) + 2\ H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$**
B. ◯ $Mg(s) + 2\ HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2(g)$
C. ◯ $Mg(s) + 2\ H^+ + 2\ NO_3^-(aq) \rightarrow Mg^{2+}(aq) + 2\ NO_3^-(aq) + H_2(g)$
D. ◯ $Mg(s) + HNO_3\ (aq) \rightarrow Mg(NO_3)_2(aq) + H_2(g)$
E. ◯ $Mg(s) + H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$

FREE-RESPONSE QUESTION (10 points)

1. The reaction below was recently reported in *Organic Syntheses*.



A balanced chemical equation for this reaction is below.

$C_6H_{12}O_6 + 2\ C_{22}H_{21}O_2P \rightarrow 2\ C_5H_6O_3 + 2\ C_2H_6O + 2$ TPPO

A. **3 points.** Use the chemical formulas and coefficients provided to determine the chemical formula of TPPO. What is the formula of TPPO?

$C_{18}H_{15}PO$

B. **2 points.** What natural law enables the determination of this chemical formula from the information given?

The law of conservation of mass

2. In one run of this reaction, 10.0 grams of $C_6H_{12}O_6$ were combined with 46.41 grams of $C_{22}H_{21}O_2P$. The reaction yielded 4.31 grams of $C_5H_6O_3$ product.

A. **2 points.** Which reactant is limiting in this reaction? Bubble in your choice.

○ $C_6H_{12}O_6$ ○ $C_{22}H_{21}O_2P$

B. **3 points.** Determine the percent yield of the reaction, filling in the boxes below.

Theoretical yield of $C_5H_6O_3$ (grams)	**12.6 g**
Actual yield of $C_5H_6O_3$ (grams)	**4.31 g**
Percent yield of $C_5H_6O_3$ (%)	**34.2% or 34%**